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MILITARY SPECIFICATION

CLOTH, NYLON, POLYCHLOROPRENE-COATED

(FOR PNEUMATIC FLOATING EQUIPMENT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers uncured and cured polychloroprene-coated nylon cloth for use as specified in 6.1.

1.2 Classification. The coated cloth shall be of the following types and classes, as specified (see 6.2):

Type 3 - Single-ply cloth, 11.5 oz/sq yd (max).
Type 4 - Single-ply cloth, 31.0 oz/sq yd (max).
Type 6 - Single-ply cloth, 44.0 oz/sq yd (max).
Type 7 - Single-ply cloth, 60.0 oz/sq yd (max).
Type 10 - Pile cloth, 40.5 oz/yd (max).
Type 11 - Single-ply cloth, 14.0 oz/sq yd (max).

Each of the above types of coated cloth is further classified according to extent of low temperature resistance as follows:

Class A - crack-resistant to -28.9° C (-20° F).
Class B - crack-resistant to -40° C (-40° F).
Class C - crack-resistant to -54° C (-65° F).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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specified in the solicitation, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods.

MILITARY

MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1487	- Glossary of Cloth Coating Imperfections.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D297	- Rubber Products - Chemical Analysis.
ASTM D412	- Rubber Properties In Tension.
ASTM D573	- Rubber-Deterioration In An Air Oven.
ASTM D750	- Rubber-Deterioration In Carbon-Arc or Weathering Apparatus.
ASTM D792	- Specific Gravity and Density of Plastics by Displacement, Tests for.
ASTM D3767	- Rubber-Measurement of Dimensions.
ASTM G23	- Operating Light and Water Exposure Apparatus (Carbon Arc) for Exposure of Nonmetallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. Unless otherwise specified (see 6.2), the coated cloth shall be uncured polychloroprene-coated nylon cloth; when specified (see 6.2),

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the cloth shall be cured. Coatings shall be applied by either the spreader or calender process, except for type 10 materials, wherein only the spreader process shall be used.

3.1.1 First article. A sample shall be subjected to first article inspection (see 4.3).

3.2 Materials.

3.2.1 Uncoated nylon cloth. The uncoated nylon cloth shall be pure finish, unbleached and shall have a pH (acidity factor) between 6 and 8; the nylon cloth shall be high-tenacity, improved heat- and light-resistant, heat-set, and may be scoured. The total chloroform-soluble and water-soluble material content of the finished cloth shall not exceed 1.0 percent when tested as specified in 4.4.2.2. Physical properties of the nylon cloth, after heat-setting and shrinking, shall conform to table I for the applicable class.

3.2.2 Primer. The nylon cloth may be impregnated with a primer to insure adhesive bond to the coating compound.

3.2.3 Coating compound. The coating compound, as prepared or vulcanized (cured), shall contain not less than 50 percent by volume of chloroprene. The coating compound shall contain no materials which are injurious to nylon cloth, which will result in a waxy finish, or which are water soluble. A compatible coloring agent shall be added to make the finished color black. After vulcanization, physical properties of the coating compound shall conform to table III.

3.2.4 Coated cloth. The primed nylon cloth (see 3.2.1 and 3.2.2) shall be coated with the compound specified in 3.2.3. The coating thickness, as specified in table II, shall be exclusive of the primer coat thickness. Physical properties of the coated cloth after vulcanization shall conform to table II.

3.3 Length and width. The length and width of the cured coated cloth shall be as specified (see 6.2).

3.4 Workmanship. The nylon cloth and the coated nylon cloth shall contain none of the following defects:

Nylon cloth

a. Any hole, cut or tear.
b. Weaving imperfections such as loose threads and slubs.
c. Abrasion marks.
d. Chronic wrinkles or folds that affect coating.
e. Oily spots that affect coating.
f. Any other imperfections affecting strength, coating adhesion or coating thickness.

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Coated cloth

a. Any hole, cut or tear.
b. Any blister or soft lump.
c. Hard lumps exceeding 0.25 square inch in area.
d. Pinholes.
e. Chronic wrinkles or folds resulting in doubling or adhesion of surfaces.
f. Imbedded foreign matter which, upon removal, leaves coating thickness not as specified.
g. Abrasion marks affecting coating thickness.

3.4.1 Defects. Defects cited in 3.4 are defined and referenced in MIL-STD-1487.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. Each linear yard of nylon cloth and coated cloth to be subjected to first article testing shall be examined as specified in 4.5.1.

4.3.1.1 Nylon cloth. Nylon cloth containing defect shall have the defective area marked or tagged by a method that remains detectable after coating. Nylon cloth with marked defects may be coated; after coating, the marked defective areas shall be rejected.

4.3.1.2 Coated cloth. Yardage containing one or more defects shall be cause for rejection.

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4.3.1.3 Coating compound. Specimens of the coating compound to be subjected to first article testing shall be cured in the form of 6 x 6 x .080 inch test slabs under the conditions of time and temperature normally used to cure fabricated end items. Tests sheets shall have no air bubbles, irregular surfaces or other visible defects.

4.3.2 Tests.

4.3.2.1 Nylon cloth and coated nylon cloth. The nylon cloth and coated nylon cloth shall be subjected to all tests specified in table IV. If applicable, the chloroform extractable matter test of 4.5.2.1.1 shall be conducted on the nylon cloth. Non-conformance to tables I or II shall constitute failure of this test and rejection of the first article lot.

4.3.2.2 Coating compound. Samples of the cured coating compound shall be subjected to all tests specified in table III. Non-conformance to table III shall constitute failure of this test and rejection of the first article lot.

4.4 Quality conformance inspection.

4.4.1 Examination. Each linear yard of nylon cloth and coated cloth shall be examined as specified in 4.5.1.

4.4.1.1 Nylon cloth. Nylon cloth containing defects shall have defective area marked or tagged by a method that remains detectable after coating. Nylon cloth with marked defects may be coated; after coating, the marked defective areas shall be rejected.

4.4.1.2 Coated cloth. Yardage containing one or more defects shall be cause for rejection.

4.4.1.3 Coating compound. Preparation and examination procedures as specified in 4.3.1.3 shall be applicable to samples of coating compound prepared for quality conformance inspection.

4.4.2 Tests.

4.4.2.1 Coating compound. Samples of coating compound shall be furnished from each batch or lot of 1000 pounds or less. Samples shall be tested as specified in 4.5.2.3. Failure of any test shall be cause for rejection of the batch and of the yardage of coated cloth represented by the sample.

4.4.2.2 Nylon cloth and coated nylon cloth. Samples of nylon cloth and coated nylon cloth shall be furnished from each 500 linear yards or less. Samples shall be tested as specified in 4.5.2.2. Failure of any test shall be cause for rejection of the yardage of cloth represented by the sample.

4.5 Inspection procedure.

4.5.1 Examination. The nylon cloth and both sides of the coated cloth shall be examined for the following defects:

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Nylon cloth

101. Weave not as specified.
102. Workmanship not as specified.

Coated cloth

103. Color or finish not as specified.
104. Length or width of the cured coated cloth not as specified.
105. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Nylon cloth.

4.5.2.1.1 Chloroform extractable matter. A sample weighing approximately 10 grams shall be dried to constant weight at 105° to 110° C (221° F to 230° F) to obtain the dry weight. The sample shall be extracted in a Soxhlet apparatus for at least 20 extractions with chloroform. The chloroform shall be evaporated from the extract and the residue dried to constant weight at 105° C to 110° C (221° F to 230°F). The result of the analysis shall be obtained by the following calculation:

$$\text{Percent chloroform extractable matter} = \frac{\text{Residue X 100}}{\text{Weight of dry sample}}$$

(Reference FED-STD-191, method 2611)

Nonconformance to the requirement of 3.2.1 shall constitute failure and rejection of the lot of uncoated nylon cloth.

4.5.2.2 Nylon cloth and coated nylon cloth. The nylon cloth and coated nylon cloth shall be subjected to the tests specified in table IV except that test procedures 12 (crease reisitance at low temeprature), 13 (oven aging), and 14 (accelerated weathering) are not required for quality conformance inspection. Nonconformance to the applicable requirements of tables I and II shall constitute failure of this test.

4.5.2.3 Coating compound. The coating compound shall be tested as specified in table III, except that the oven aging and accelerated aging procedures are not required for quality conformance inspection. Nonconformance to the applicable requirements of table III shall consitute failure of this test.

4.6 Inspection of packaging. Each shipping container of cloth shall be examined for the following defects as applicable:

a. Precautions not taken to prevent blocking of coated cloth.
b. Marking not in accordance with MIL-STD-1188.
c. Special marking not placed on shipping containers when applicable.

Any defects shall be corrected before shipment is made.

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5. PACKAGING

5.1 Preservation.

5.1.1 Level A and B. Not applicable (see 6.3).

5.1.2 Commercial. Uncured and cured cloth shall be preserved in accordance with the contractor's recommended procedures and MIL-STD-1188. Special precautions shall be taken to prevent blocking of the coated cloth.

5.2 Packing.

5.2.1 Levels A and B. Not applicable (see 6.3).

5.2.2 Commercial. Both uncured and cured cloth shall be packed in containers in accordance with MIL-STD-1188. Quantities per container of each type of cloth shall be compatible with the production, handling, and storage capabilities of the receiving organization.

5.3 Marking. Marking of commerical packaged cloth shall be in accordance with MIL-STD-1188. In addition, special precautionary marking related to shelf life and exposure to high temperatures shall be applied to shipping containers when applicable.

6. NOTES

6.1 Intended use. This specification covers uncured polychloroprene coated nylon cloth which will be vulcanized (cured) during fabrication of pneumatic floating equipment. The cured coated cloth is intended for use in repair kits.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of cloth required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. When the coated cloth shall be cured (see 3.1).
e. Length and width of vulcanized coated cloth (see 3.3).

6.3 Preservation levels. It is anticipated that the coated and uncoated cloth will not require shipment other than from the contractor to the ponton fabricator or repair kit contractor and therefore will not encounter the hazards defined for levels A and B, overseas shipment and indeterminate storage.

Custodian:
Army - ME
Air Force - 20

Review activity:
Army - MD, GL

User activity:
Air Force - 11, 82

Preparing activity:
Army - ME

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TABLE I. Properties of uncoated nylon cloth types 3, 4, 6, 7 and 11, all classes.

Property	Requirements					Test Number, Table IV
	Type 3	Type 4	Type 6	Type 7	Type 11	
Weight (oz/sq yd) max	3.50	5.30	9.0	13.30	4.3	1
Weave	Plain or 2x2 basket	Plain, except with 2 rip-stop yarns per inch, or 2x2 basket	2x2 basket	2x2 basket	Plain	Visual
Count						
Warp (yarns/in) min	46	42	33	40	78	3
Fill (yarns/in) min	36	42	33	40	57	
Breaking strength						
Warp (lb/in) min	170	350	400	650	225	4, 6 1/
Fill (lb/in) min	150	335	400	650	180	
Tearing strength						
Warp (lb) min	12	50	80	100	15	7
Fill (lb) min	12	50	80	100	15	

1/ Breaking strength shall be determined by the grab method (4) for type 4 only. The ravel strip method (6) shall be used for all other types.

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TABLE I. Type 10 - uncoated nylon pile cloth, all classes. (Cont'd)

Construction: Cloth shall consist of two face cloths of plain weave connected by pile threads. The pile thread shall be continuous, weaving between the face cloth in planes parallel to the warp threads of the face cloths. At each junction with the face cloth, the pile thread shall go over not less than two and under not less than one fill thread. When pile yarns are fully extended, the distance between face cloths shall be not less than 2.25 inches and not more than 3.0 inches.

Property	Requirement	Test Number, Table IV
Weight (oz/sq yd) (2 face cloths and pile thread) max	8.5	1
Pile threads		
Denier (grams/9.0 km) min	210	
Breaking strength (lb/yarn) min	3	8
Count (yarns/sq in) min	30	3
Face cloth		
Count		3
Warp (yarns/in) min	47	3
Fill (yarns/in) min	47	3
Yarn denier		
Warp - 2 ply yarn, each (grams/9.0 km) min	70	
Fill - single ply (grams/9.0 km) min	210	
Face cloth (pile threads completely removed)		
Weight (oz/sq yd) max	3.0	1
Breaking strength		
Warp (lb) min	100	6
Fill (lb) min	150	6
Tearing strength		
Warp (lb) min	10	7
Fill (lb) min	15	7

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TABLE II. Properties of vulcanized coated cloth - types 3, 4, 6, 7, 10 and 11, all classes.

Property	Requirements						Test Number, Table IV
	Type 3	Type 4	Type 6	Type 7	Type 10	Type 11	
Coating thickness (mils) Each side or exterior surface	3.3	6.0	10.0	16.0	11.0	3.5	15
Total weight (oz/sq yd) max	11.5	31.0	44.0	60	40.5	14.0	1
Porosity - (air retention)	-	No leaks	No leaks	No leaks	No leaks	No leaks	16
Low-temperature resistance at: -28° C (-20° F) -40° C (-40° F) or -54° C (-65° F)	No cracking or flaking	No cracking or flaking	No cracking or flaking	No cracking or flaking	No cracking or flaking	No cracking or flaking	11
Coating adhesion (lb/in) min	8	20	20	20	10	12	9
Water resistance coating adhesion retained after exposure to water at 60° C(140° F) for 96 hours - (lb/in) min.	5	12	12	12	6	7.5	14, 9

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TABLE II. Properties of vulcanized coated cloth - types 3, 4, 6, 7, 10 and 11, all classes. (Cont'd)

Property	Requirements												Test Number, Table IV
	Type 3		Type 4		Type 6		Type 7		Type 10		Type 11		
	Warp	Fill	Warp	Fill	Warp	Fill	Warp	Fill	Warp	Fill	Warp	Fill	
Tearing strength (lb) min	8	8	25	25	50	50	70	70	7	10	8	8	7
Breaking strength (lb) min	153	135	350	335	400	400	650	650	100	150	225	180	4
After oven aging at 100° C (212° F) for 96 hr, breaking strength retained (percent) min	85	85	85	85	85	85	85	85	-	-	85	85	12, 4, 5
After accelerated weathering for 100 hr, breaking strength retained (percent) min.	85	85	85	85	85	85	85	85	-	-	85	85	13, 4, 5

Notes: Breaking strength shall be determined by the grab method (4) for type 4 only. The cut step method (5) shall be used for all other classes.
Low temperature resistance, oven aging, and accelerated weathering tests required for preproduction inspection only.

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TABLE III. Properties of vulcanized coating compound.1/

Property	Requirement	ASTM Method
Specific gravity, max	1.650	D792
Coating compound identification	Polychloroprene	D297
Chloroprene content (percent by volume) min	50	D297, par 53
Tensile strength (psi) min	1800	D412
Ultimate elongation (percent) min	500	D412
After oven aging at 100° C (212° F) for 94 hr Tensile strength retained (percent) min	- 85	D573 D412
Ultimate elongation retained (percent) min	85	D412
After accelerated weathering at 10 percent elongation for 100 hr	-	D750, G25 2/
Tensile strength retained (percent)	85	D412
Ultimate elongation retained (percent) min	85	D412

1/ If more than one type of compound is used for coating, all types of compounds must meet the above requirements.

2/ Accelerated weathering (coating compound) - The light aging method specified in ASTM D750 shall be followed with the following exceptions. After buffing, all specimens shall be stretched to 10 percent elongation within 3 hours. The stretched samples shall be aged in a circulating air oven for 16 to 24 hours at 37.8° C (100° F), ±2° C (5° F), immediately after racking to facilitate blooming of protective waxes. The exposure in the light aging unit (type E of ASTM G23), shall be started within 1 hour after removal from the 37.8° C (100° F) oven. Exposure for 100 hours shall be made at 10 percent elongation using Eveready "Sunshine" carbons or aproved equal. Four of the 8 "Corex D" filters shall be removed leaving each alternate glass filter in place.

Note: Oven aging and accelerated weathering tests required for preproduction inspection only.

TABLE IV. Test procedure - uncoated cloth and vulcanized coated cloth.

Test No.	Characteristic	FED-STD-191	Number of specimens	Results reported as:
1	Weight	5041 1/	5	Aver. of 5 trials to nearest 0.1 oz/sq yd.
2	Thickness	5030 1/	5	Aver. of 5 trials to nearest mil.
3	Yarn count	5050 1/, 2/	5	Aver. of 5 trials to nearest whole number.
4	Breaking strength (grab)	5100 1/	5	Aver. of 5 trials to nearest pound.

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TABLE IV. Test procedure - uncoated cloth and vulcanized coated cloth.[1/] (Cont'd)

Test No.	Characteristic	FED-STD-191	Number of specimens	Results reported as:
5	Breaking strength (cut strip)	5102[1/,4/]	5	Aver. of 5 trials to nearest pound per inch.
6	Breaking strength (ravel strip)	5104[1/,3/]	5	Aver. of 5 trials to nearest pound per inch.
7	Tearing strength (tongue)	5134[1/,5/,3/]	5	Aver. of 5 trials to nearest 0.1 pound.
8	Breaking strength (yarn)	4102[1/,6/]	5	Aver. of 5 trials to nearest 0.1 pound.
9	Coating adhesion	5970[1/,7/]	3	Aver. of 3 trials to nearest 0.1 pound per 2 inches of width.
10	Ply adhesion	5970[1/,8/]	3	Aver. of 3 trials to nearest 0.1 pound per 2 inches of width.
11	Crease resistance to low temperature	5874[1/,12/]	3	Cracking or flaking of coating.
12	Oven aging	5850[1/,9/]	5	Aver. of 5 trials. Report retention of average breaking strength of unaged specimens to nearest 1.0 percent.
13	Accelerated weathering	5804[1/,10/]	5	Aver. of 5 trials. Report retention of average breaking strenth of unaged specimens to nearest 1.0 percent.
14	Water resistance[13/]	-	3	Aver. of 3 trials to nearest 0.1 pound per 2 inches of width.
15	Thickness (coated fabric)	2011[11/]	5	Aver. of 5 trials to nearest mil.
16	Porosity (air retention)	[14/]	5	Leak or not leak.
17	Identification of woven cloth material	1530	3	Presence or absence of nylon filaments.

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1/ In tests of multiple-ply coated cloth, biased-laminated, the direction (warp or fill) of the heavier or straight ply shall determine the corresponding direction of the laminate.

2/ a. Multiple yarns such as double yarns of rip-stop or basket weave shall each be counted. For example, a double yarn shall be counted as two yarns.
b. In counting tie or pile yarns of pile cloth, the count in yarns per square inch shall be the tie threads in the warp direction multiplied by the tie threads in the fill direction.

3/ a. Specimens of the face cloths of uncoated pile fabric for breaking and tearing strength tests shall be prepared after all pile threads have been cut as close to the surface of the face cloth as practicable. Remnants of the pile threads shall not be removed.
b. Specimens of the face cloth of coated pile fabric for breaking and tearing strength tests shall be prepared after all pile threads have been cut as close to the face cloth as possible without damaging the face cloth.

4/ For determination of breaking strength by cut-strip method, extreme care shall be taken to cut the specimens parallel to the yarns to be tested so that all yarns of the specimen extend for the full length of the specimen. To do this, specimen may be torn (if an easily torn coated fabric) or a few threads cut wider than 1.0 inch and the extra threads then raveled from opposite edges.

5/ a. For tearing strength of uncoated or single-ply coated cloth, machine jaw separation rate shall be the standard 12 inches per minute.
b. For multiple-ply, laminated, coated cloths, the rate shall be 2.0 inches per minute. If plies delaminate, the lighter or bias ply shall be cut back to the line of tear in the heavier ply.

6/ Specimens of tie or pile threads of uncoated pile fabric shall be prepared by unraveling a continuous piece long enough for the test (ordinarily not less than 24 inches). The face cloths from which the pile thread has been thus removed should not be used for strength tests since the warp and fill threads may have been damaged.

7/ Determinations shall be made on each face of the coated fabric or laminate by method 5970 with the following exceptions:

a. In lieu of the method described in 4.1 through 4.3 of method 5970, specimens shall be made by bonding two uncured pieces of the coated fabric together (face-to-face or back-to-back) using the solvent or adhesive system to be used in making seams in the pneumatic floating equipment. The specimens shall then be cured in a manner identical to that of the fabricated item.
b. The rate of jaw separation shall be 2.0 inches per minute.
c. If the coating separates from both pieces forming a web between pieces, cut the web back to the cloth to which the least rubber has adhered.

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8/ Determination shall be made of the adhesion between each pair of fabrics in a laminate. The test will be performed by method 5970 as modified by Note (7b) and (7c). Specimens shall be cut from the laminate, and the plies shall be separated manually at one end for a sufficient distance to enable attachment to the jaws of the test machine. A little benzene or toluene may be used to assist in the separation; but if benzene or toluene is used, the first part of the adhesion curve shall be disregarded as it may have been affected by the solvent.

9/ Oven aging: specimens and test method shall be the same as that used in determining property before aging. Specimens shall be aged in oven at 100° C (212° F) for 94 hours.

10/ Accelerated weathering: Alternate "Corex D" filters shall be removed from the weatherometer. Specimens and test method shall be the same as that used in determining property before aging. Specimens shall be exposed to radiation for 100 hours.

11/ a. ASTM D3767, 8.1, method A.
b. Coating thickness shall be determined as follows:

Coating A = $T_c - T_s$.

Coating B = $T_s - T_u$.

Where T_c = thickness of coated cloth.

T_u = thickness of uncoated cloth.

T_s = thickness of cloth and coating on one side.

To ascertain the thickness of the cloth and coating on one side, the coating on one side shall be peeled by using a volatile solvent such as benzene, and the specimen shall be dried thoroughly.

12/ Crease resistance to low temperature: Six by six inch test specimens shall be cut from coated fabric samples; each specimen shall be subjected to two folds at 90 degrees to each other at 23° C (73° F), and then placed in a chamber at -28.9° C (-20° F), -40° C (-40° F), or 54° C (-65° F) ±1° C (2° F), between two rigid, flat, metal plates under a total load of 6 pounds. The assembly shall be supported a sufficent distance from the sides and bottom of the chamber and the assembly shall remain at -20° F, -40° F, or -65° F, ±2° F, at all times during a 96-hour conditioning period. At the end of the 96 hours at the test temperature, the specimen shall be unfolded rapidly in the test chamber, using gloves previously cooled at the test temperature for not less than 1 hour. The specimens shall be removed from the conditioning chamber and examined visually with the aid of a 7-power magnifying glass for cracking or flaking of the coating.

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13/ Water resistance test: Test specimens for coating adhesion shall be immersed in distilled water at 60° C (140° F) 140° F, ±10° C (2° F), for 96 hours, ±1 hour. The edges of the test specimens shall not be sealed. The test specimens shall then be removed and immediately placed in water at room temperature. Determination of physical properties shall be made as required. Each determination shall be completed within 3 minutes of time of removal from the water.

14/ Porosity - (air retention): Three specimens (13 inches in diameter) of the coated cloth shall be individually tested on the test jig as shown in figure 1. The specimen shall be placed, (coated side down for type 10 specimens), on the holder and the plate collar tightly bolted thereon. Care should be taken to insure a leaktight fit. Water shall be poured on top of the specimen, sufficient to keep the top of the specimen completely covered at all levels of pressure. The specimen shall be inflated to an air pressure of 10 psi for 5 minutes. The air bubbles on the cloth surface produced by air pressure closing the spaces between the cloth and coating shall be removed. The cloth shall show no signs of leakage as evidenced by continued production of air bubbles.

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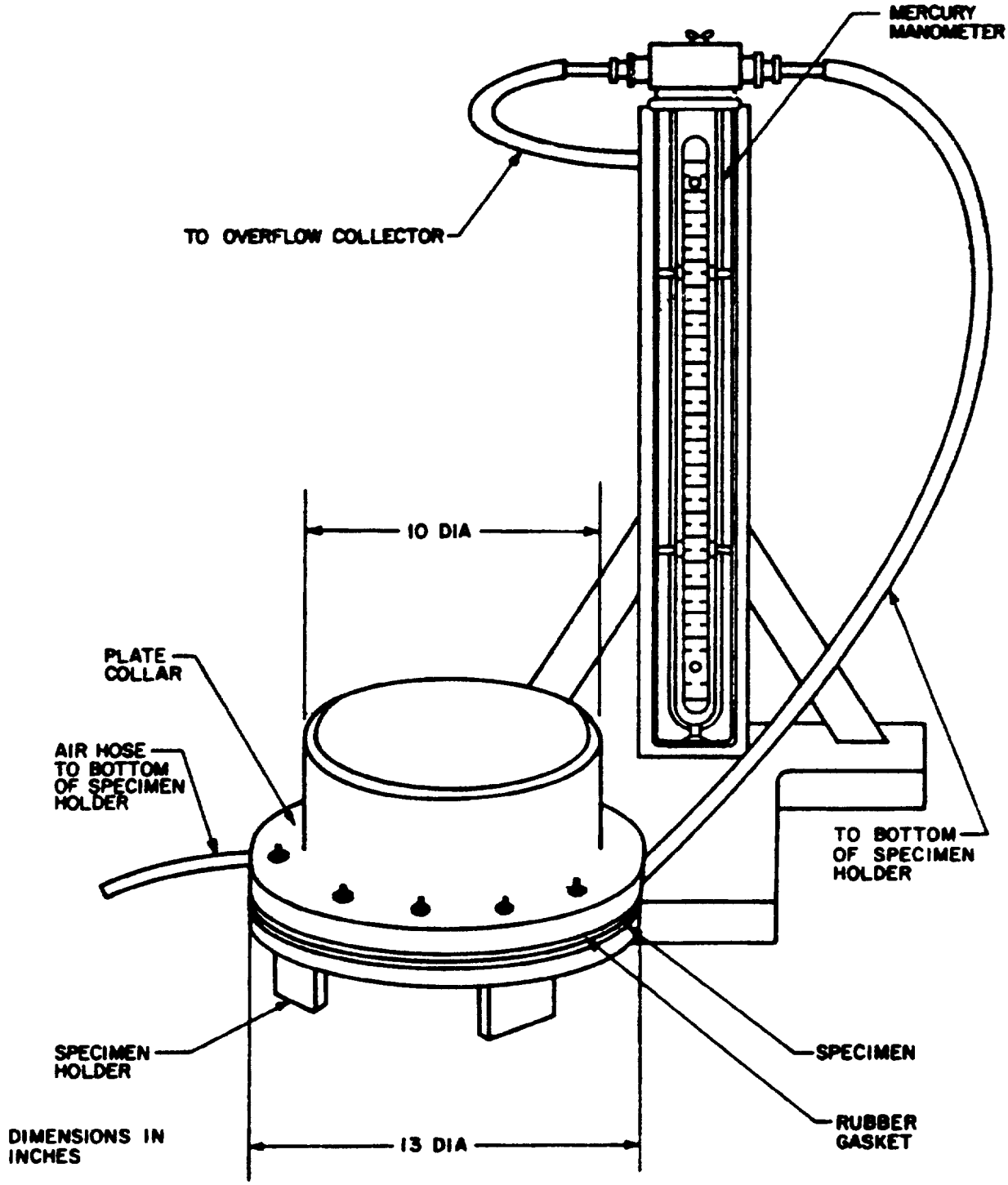


FIGURE 1. Air leakage test jig.

X-4095

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-14505D	Cloth, Nylon, Polychloroprene-Coated (for Pneumatic Floating Equipment

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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